

HOW TO PLAY WITH LIGHT

Using LDR

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In electronics, there are many ways to play with light the way you dream. If you are looking for an inexpensive solution, regular light-dependent resistors (LDRs) can also be used as optical sensors.

Light-dependent resistors based on cadmium-sulphide (CdS) have a resistance that varies inversely with the amount of light falling on them. These LDRs are known by many names, including photoresistor, photoconductive cell or simply the photocell. Usually, LDRs are available in epoxy resin and hermetical packages (Fig. 1).

You can take a few measure-

ments to determine the mathematical relationship between the LDR's resistance and lux values by following the steps given below:

1. Insert the LDR into a breadboard so that the flat top of the LDR is parallel to the ground plane
2. Connect your digital multimeter (with its knob turned to the resistance range) to the two leads of the LDR
3. Place the sensor of the lux meter (Fig. 2) near the LDR
4. While ensuring that the same amount of light falls on the LDR and sensor, note down the LDR's resistance and lux values in lux meter. Repeat this process for several different light and lux levels ranging from very dark to very bright light
5. Finally, transfer your findings to a spreadsheet and plot resistance as a function of light. Typically, the graph of LDR resistance as a function of light looks as shown in Fig. 3

CdS LDRs exhibit sensitivity curves that closely match the sensitivity of the human eye, which makes them useful in applications involving human light perception, such as headlight dimmers and intensity adjustments on information displays.

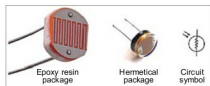


Fig. 1: LDR packages

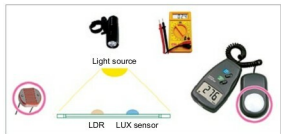


Fig. 2: Measuring the values of an LDR using lux meter

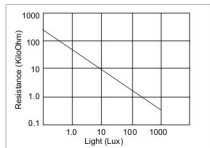


Fig. 3: Typical graph of LDR resistance as a function of light

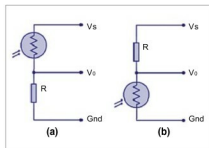


Fig. 4: Potential divider using LDR: (a) LDR at the top, (b) LDR at the bottom

LDR as a potential divider

Since the resistance of LDRs varies in response to light, usually potential divider concept is used to get a valid output for further processing by other parts of the circuit (for example, analogue-to-digital controller of a microcontroller). As shown in Fig. 4, when the LDR is at the top, we get a large V_o as the sensor has a low resistance, and when the LDR is at the bottom, we get a small V_o as the sensor has a high resistance.

Practical application of LDR: smart bulb holder

Smart bulb holder is nothing but an AC-mains-operated incandescent, compact fluorescent or LED light bulb holder with a simple dusk-to-dawn/twilight controller circuit for automatic switching of your outdoor/corridor electric light bulbs (Fig. 5). The circuit, as shown in Fig. 6, can be enclosed in an appropriate light bulb holder. It comprises two equally important parts:

PARTS LIST

Semiconductors:

- T1, T2 - BC547 npn transistor
- D1-D5 - 1N4007 rectifier diode
- ZD1 - 12V, 1W zener diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless specified):

- R1 - 820-kilo-ohm
- R2 - 15-ohm, 2W
- R3 - 1.5-kilo-ohm
- R4 - 10-kilo-ohm
- R5 - 1-kilo-ohm
- VR1 - 22-kilo-ohm potmeter
- LDR1 - 5mm photoresistor or light-dependent resistor

Capacitors:

- C1 - 1 μ F, 400V polyester
- C2 - 470 μ F, 25V electrolytic
- C3 - 22 μ F, 25V electrolytic

Miscellaneous:

- CON1, CON2 - 2-pin terminal connector
- F1 - 3A fuse with holder
- RL1 - 12V, 1C/O relay